THE COZY SCIENCE LAB · ORIGINAL STUDY SERIES

Chemistry

A Grade 7–10 study handbook with concept notes, examples, self-check explanations and labelled illustrations.

In this handbook

- Matter & Mixtures
- Atoms & the Periodic Table
- Bonding & Compounds
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Edition 1 · Lab-authored learning materials. Supporting publications are credited with each topic.
Examples are constructed for teaching; diagrams state their model and scale limits. For current Philippine hazard or weather decisions, consult the responsible agencies rather than a static study guide.

Matter & Mixtures

Matter has mass and occupies space. Particle models help explain states, mixtures and separation methods.

States and particle motion

In a solid, particles fluctuate around relatively fixed positions. Liquid particles can move past one another; gas particles are widely separated and move through their container. Heating changes particle motion and can cause a change of state.

Substances and mixtures

An element contains one atomic number; a compound has elements chemically combined in a defined composition. A mixture contains substances together without all being chemically combined into one new substance.

Choosing a separation

Separation methods use differences in properties. Filtration separates suitable insoluble particles from a fluid; evaporation or distillation can help separate some solutions. Choose a method that suits the materials and the desired product.

Key terms

Solution

A homogeneous mixture at the molecular or ionic scale.

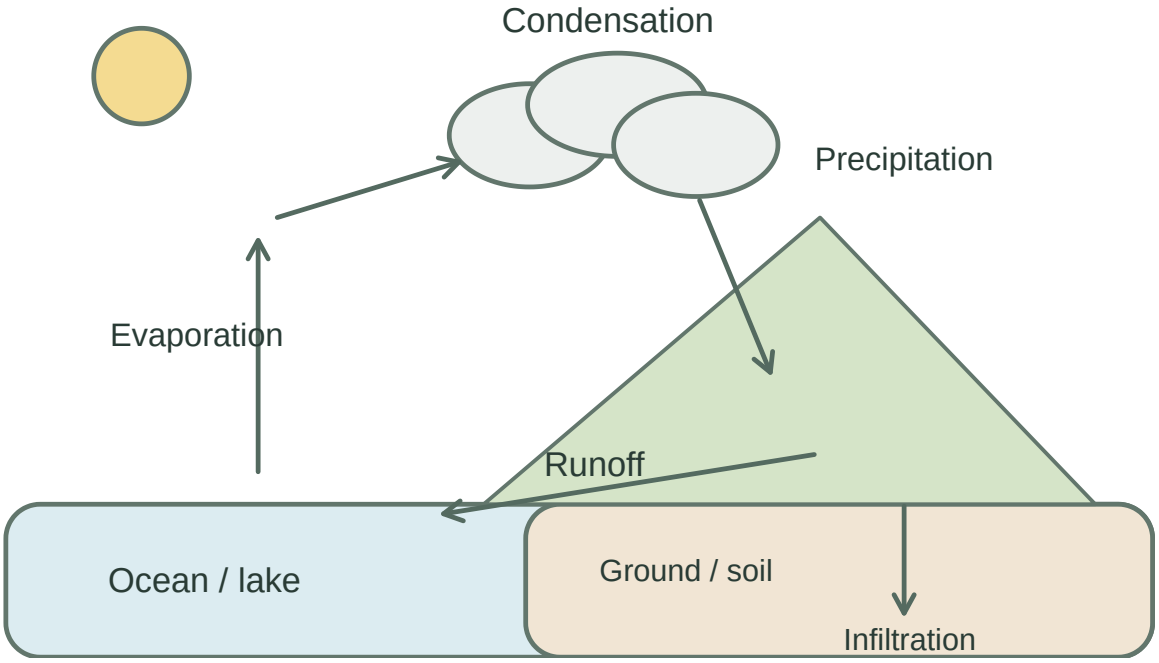
Physical change

A change that does not alter the chemical identities of the substances involved.

Salt water and a filter

Dissolved ions usually pass through an ordinary classroom filter with the water. Filtering sand from water works because the insoluble grains are large enough to be retained.

Remember: During melting, the particles themselves do not usually melt into a different kind of particle; their arrangement and motion change.



Follow a water molecule. Simplified pathways, not to scale. Water vapour is invisible; clouds contain tiny liquid droplets and/or ice particles.

Check your thinking

Why is distillation different from simply filtering a solution?

Explanation: Distillation uses differences in volatility and involves vaporization and condensation. Ordinary filtration cannot retain individual dissolved ions.

Supporting references

These original notes summarize school science concepts. They are not quotations or official curriculum pages.

OpenStax · Rice University. Chemistry 2e. https://openstax.org/details/books/chemistry-2e

Philippine Department of Education. K to 12 Science Curriculum Guide. https://www.deped.gov.ph/wp-content/uploads/2019/01/Science-CG_with-tagged-sci-equipment_revised.pdf

Atoms & the Periodic Table

Elements are organized by atomic number. Use the periodic table to connect an atom's identity with its place in a larger pattern.

Identity of an element

Atomic number is the number of protons in the nucleus. A neutral atom has the same number of electrons as protons. Electrons occupy energy states and regions of probability, not miniature planetary paths.

Isotopes and ions

Isotopes of an element have the same proton count but different neutron counts. An ion has gained or lost electrons. Changing electrons does not change the atomic number.

Reading the table

Periods are rows and numbered groups are columns. Elements in a group can show related chemical behaviour. The separately displayed lanthanide and actinide series are still part of periods 6 and 7.

Key terms

Mass number

The total protons plus neutrons for a specified isotope.

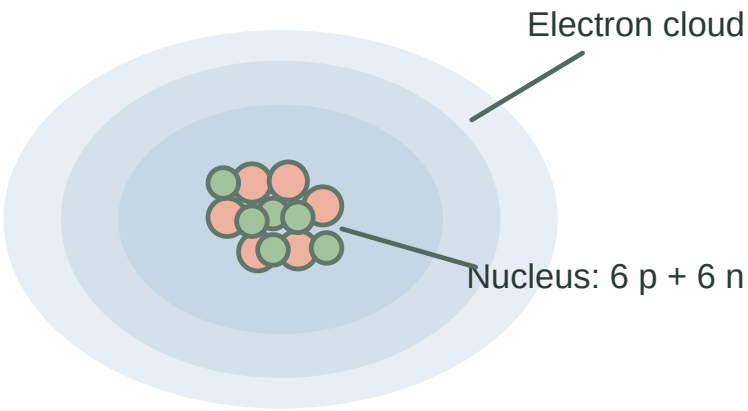
Atomic number

The proton count that defines an element.

Carbon-14

Carbon has atomic number 6. Carbon-14 therefore has 6 protons and 14 − 6 = 8 neutrons. A neutral carbon-14 atom has 6 electrons.

Remember: Mass number, atomic number and a periodic table's standard atomic weight are different quantities. Do not round a weight to invent the neutron count of an unspecified atom.



Carbon-12 · neutral atom: 6 electrons

An atom is not a tiny solar system. Conceptual model, not to scale or a calculated probability plot. Electrons do not follow the drawn edges as planetary paths.

Check your thinking

If sodium loses one electron, does it become a different element?

Explanation: No. It becomes a sodium ion with a +1 charge; its 11 protons still identify it as sodium.

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Bonding & Compounds

Bonding models explain how atoms form substances and why those substances have different properties.

Ionic bonding

Electron transfer can produce oppositely charged ions. Ionic bonding is the electrostatic attraction between ions; an ionic solid such as sodium chloride forms an extended lattice.

Covalent bonding

Covalent bonding involves shared electron density between atoms. Small covalent molecules and giant covalent networks can have very different melting points and other properties.

Metallic bonding

In a metal, mobile, delocalized electrons are associated with an array of metal centres. This model helps explain electrical conductivity and why many metals can be shaped without simply shattering.

Key terms

Ion

An atom or group of atoms with a net electric charge.

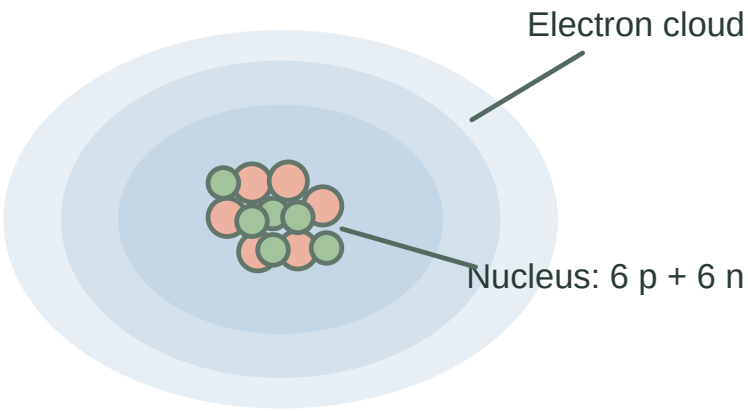
Lattice

A repeating arrangement of particles in a solid.

NaCl

In the simple ion model, sodium forms Na+ and chlorine forms Cl−. Their charges balance in a 1:1 ratio. The formula describes the ratio in a lattice, not an isolated molecule in a salt crystal.

Remember: The octet rule is a useful introductory pattern with exceptions. It does not explain every bond or every element.



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Check your thinking

Why can an ionic substance's solution conduct electricity?

Explanation: Dissolved ions can move and carry charge. In a rigid ionic crystal, those ions are not free to move through the material in the same way.

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Acids, Bases & pH

Acid–base ideas help describe aqueous solutions. Compare measurements carefully and distinguish strength from concentration.

Acidic, neutral and basic

For ordinary aqueous solutions near 25 °C, pH below 7 is acidic, pH 7 is neutral and pH above 7 is basic. Neutral pH depends on temperature; 0–14 is a common teaching range, not an absolute limit.

A logarithmic scale

A difference of one pH unit corresponds approximately to a tenfold difference in hydrogen-ion concentration in dilute solutions. A small-looking numerical change can therefore represent a large chemical difference.

Strength and amount

Acid strength concerns the extent of ionization under specified conditions. Concentration concerns how much solute is present in a volume. A strong acid can be dilute and a weak acid can be concentrated.

Key terms

Indicator

A substance whose observable change can help identify a chemical condition such as pH.

Neutralization

An acid–base reaction; the final pH depends on the reactants and their amounts.

pH 3 and pH 4

Under the dilute-solution approximation, a pH 3 solution has about ten times the hydrogen-ion concentration of a pH 4 solution.

Remember: A final solution is not automatically pH 7 after any acid and base are mixed. Never identify an unknown solution by tasting it.

Check your thinking

Does 'weak acid' mean there is only a small amount of acid?

Explanation: No. Weak describes ionization behaviour; dilute describes a low concentration.

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Reactions & Balancing

Chemical reactions rearrange atoms. Equations and energy changes help us describe the rearrangement.

Conserving atoms

Chemical equations show reactants and products. Balance equations by changing coefficients, keeping the formulas intact. In ordinary chemical reactions in a closed system, total mass is conserved.

Energy transfer

An exothermic process transfers energy to its surroundings; an endothermic one takes energy from them. Breaking bonds requires energy, while forming bonds releases energy. The overall balance determines the net change.

Rate and catalysts

Reaction rates depend on conditions such as temperature, concentration and exposed surface. A catalyst provides a different reaction pathway and is regenerated overall; it does not supply unlimited energy.

Key terms

Coefficient

A number multiplying a whole chemical formula in an equation.

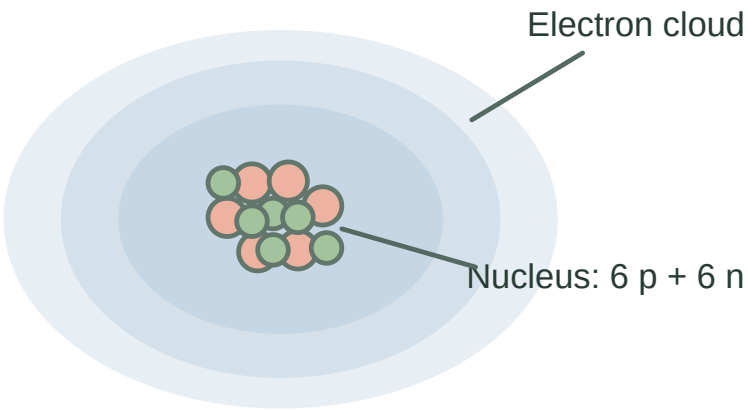
Reactant

A substance consumed in a chemical reaction.

Balancing water formation

2 H2 + O2 → 2 H2O gives four hydrogen atoms and two oxygen atoms on each side. Changing H2O to H2O2 would describe a different substance, not a balancing step.

Remember: If gas escapes an open container, a balance may show less mass remaining inside. That does not mean the atoms were destroyed.



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Check your thinking

Can a catalyst create more atoms than the reactants contained?

Explanation: No. A catalyst can affect a reaction pathway and rate, but atom conservation still applies.

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Gas Laws

Gas models connect the motion of particles with pressure, volume and temperature.

Pressure from collisions

Gas particles move and collide with the walls of a container. Pressure is force per unit area. The ideal-gas model makes simplifying assumptions that work best away from extreme conditions.

Controlled comparisons

For a fixed amount of gas at constant temperature, pressure varies inversely with volume. At constant volume, increasing absolute temperature increases pressure in the ideal model.

Use absolute temperature

Gas-law temperature ratios use kelvin. Convert from degrees Celsius using T in K = temperature in °C + 273.15. Stating what remains fixed is part of using a gas law correctly.

Key terms

Pressure

Force divided by area; the SI unit is the pascal.

Kelvin

The SI unit of thermodynamic temperature, symbol K without a degree sign.

A volume change

A fixed amount of gas at 100 kPa occupies 2 L. If temperature stays constant and volume becomes 4 L, Boyle's-law reasoning gives 50 kPa.

Remember: Doubling a Celsius reading does not generally double absolute temperature. Gas particles do not themselves swell to fill a larger container.

Check your thinking

What must stay fixed when using the simple inverse pressure–volume relationship?

Explanation: The amount of gas and its temperature must remain constant, within the limits of the model.

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